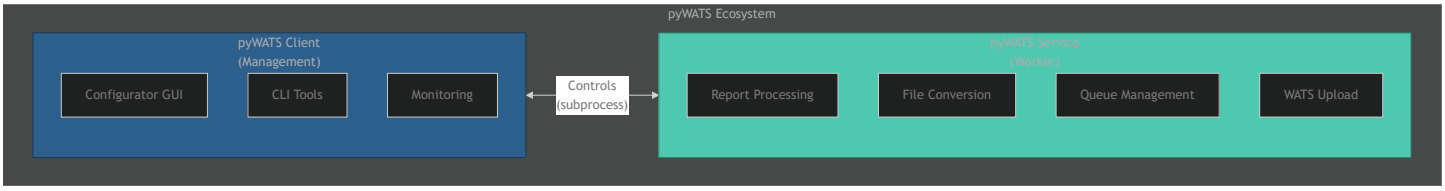
pyWATS Client & Service Architecture Guide

Last Updated: February 14, 2026
Version: 0.2.0+

Overview

The pyWATS system uses a **decoupled architecture** where the **Service** (background worker) runs independently from the **Client** (configuration and management tools). This separation ensures reliability, scalability, and flexibility in manufacturing test environments.



pyWATS Service

Purpose

The **pyWATS Service** is a **background worker process** that handles the core mission-critical operations:

1. **Test Report Processing** - Monitors test stations for new reports
2. **File Format Conversion** - Runs converters on proprietary test formats
3. **Queue Management** - Manages report upload queues
4. **WATS Integration** - Uploads test data to Teradyne WATS server
5. **Automated Operation** - Runs 24/7 without user intervention

Key Characteristics

- ✅ **Headless** - Runs without GUI (can run as Windows Service)
- ✅ **Independent** - Survives configurator/GUI crashes
- ✅ **Persistent** - Continues running until explicitly stopped
- ✅ **Instance-based** - Multiple services can run simultaneously (multi-instance)
- ✅ **Asynchronous** - Uses asyncio for concurrent operations
- ✅ **Resilient** - Auto-recovers from network failures

Responsibilities

Responsibility	Description
Report Watching	Monitors configured directories for new test reports
File Conversion	Executes converters to transform proprietary formats
Upload Management	Queues and uploads reports to WATS server
Error Handling	Retries failed uploads, logs errors
Health Monitoring	Tracks converter health, queue depth, success rates
Lock File Management	Maintains service.lock for process tracking

Implementation

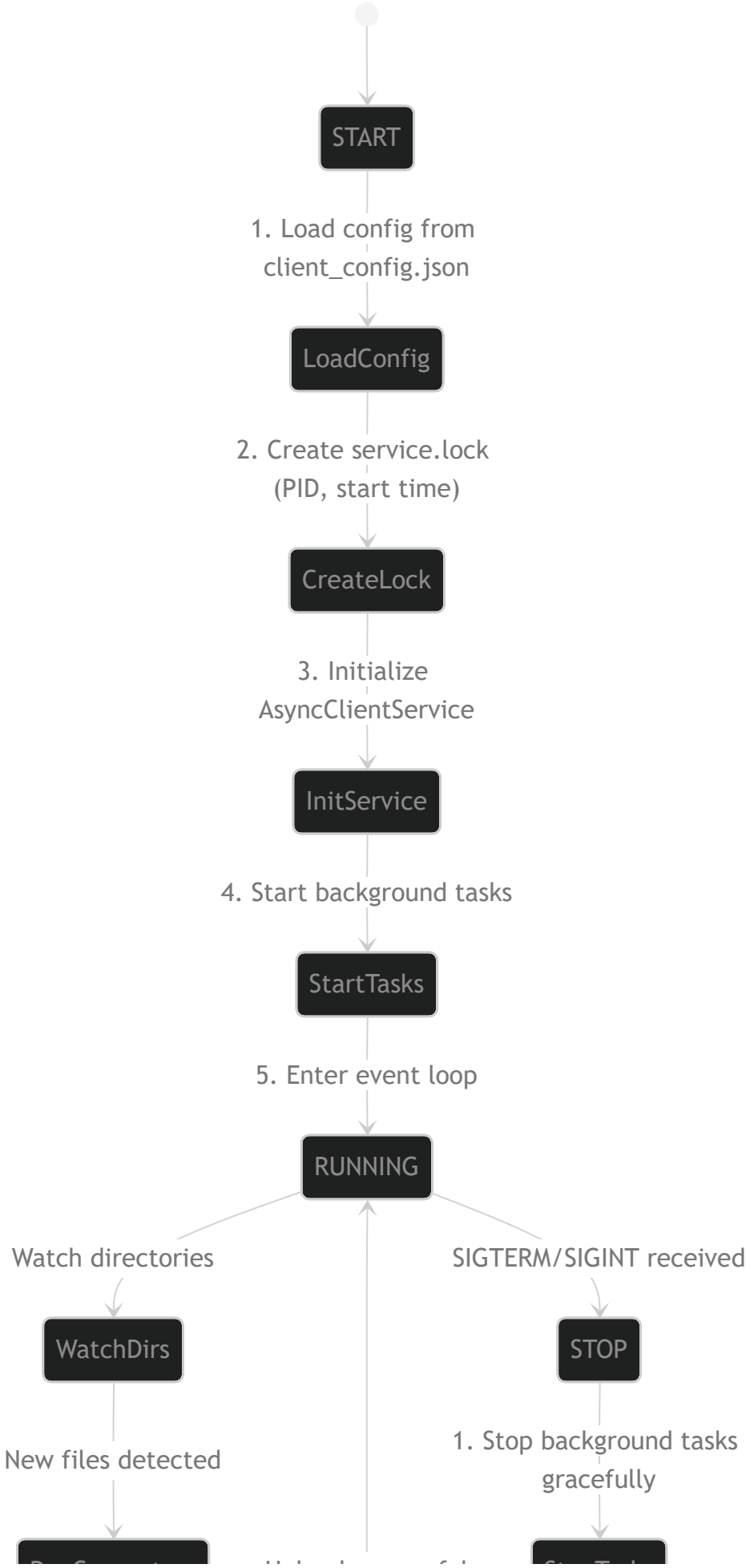
Core Class: `AsyncClientService` (src/pywats_client/service/async_client_service.py)

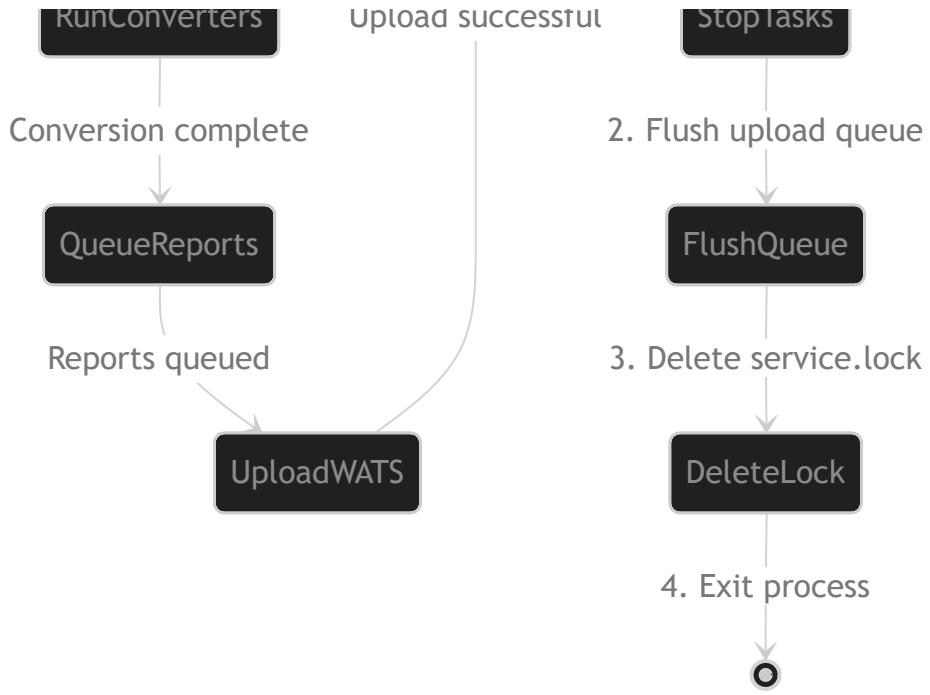
Launcher: `run_service.py` (standalone, no GUI)

Instance Data:

```
C:\ProgramData\pyWATS\instances\{instance_id}\
├── client_config.json     # Configuration
├── service.lock           # Process tracking (PID, uptime)
├── queue/                 # Upload queue
├── logs/                  # Service logs
├── reports/               # Local report archive
└── converters/            # Converter executables
```

Service Lifecycle


START
1. Load config from client_config.json
LoadConfig
2. Create service.lock (PID, start time)
CreateLock
3. Initialize AsyncClientService
InitService
4. Start background tasks
StartTasks
5. Enter event loop
RUNNING
Watch directories
WatchDirs
New files detected
SIGTERM/SIGINT received
STOP
1. Stop background tasks gracefully



pyWATS Client

Purpose

The **pyWATS Client** provides **management and monitoring tools** for pyWATS services:

1. **Configuration Management** - Edit service settings, converters, connections
2. **Service Control** - Start/stop/restart services
3. **Status Monitoring** - View service health, uptime, statistics
4. **Interactive Testing** - Manual report uploads, connection testing
5. **Setup & Deployment** - Initial configuration, multi-instance setup

Key Characteristics

- ✅ **Interactive** - User-facing GUI and CLI tools
- ✅ **Optional** - Service runs independently (client not required for operation)
- ✅ **Controller** - Launches/stops services as subprocesses
- ✅ **Transient** - Can be started/stopped without affecting running services
- ✅ **Multi-Instance Aware** - Manages multiple service instances

Responsibilities

Responsibility	Description
Configuration Editing	Modify service settings (server URL, tokens, converters)
Service Management	Start/stop/restart service processes
Status Display	Show service health, uptime, queue depth, converter status
Manual Operations	Upload reports manually, test connections
Initial Setup	First-time configuration, instance creation
Troubleshooting	View logs, check errors, diagnose issues

Implementation

GUI Application: `Configurator` (src/pywats_ui/apps/configurator/main_window.py)

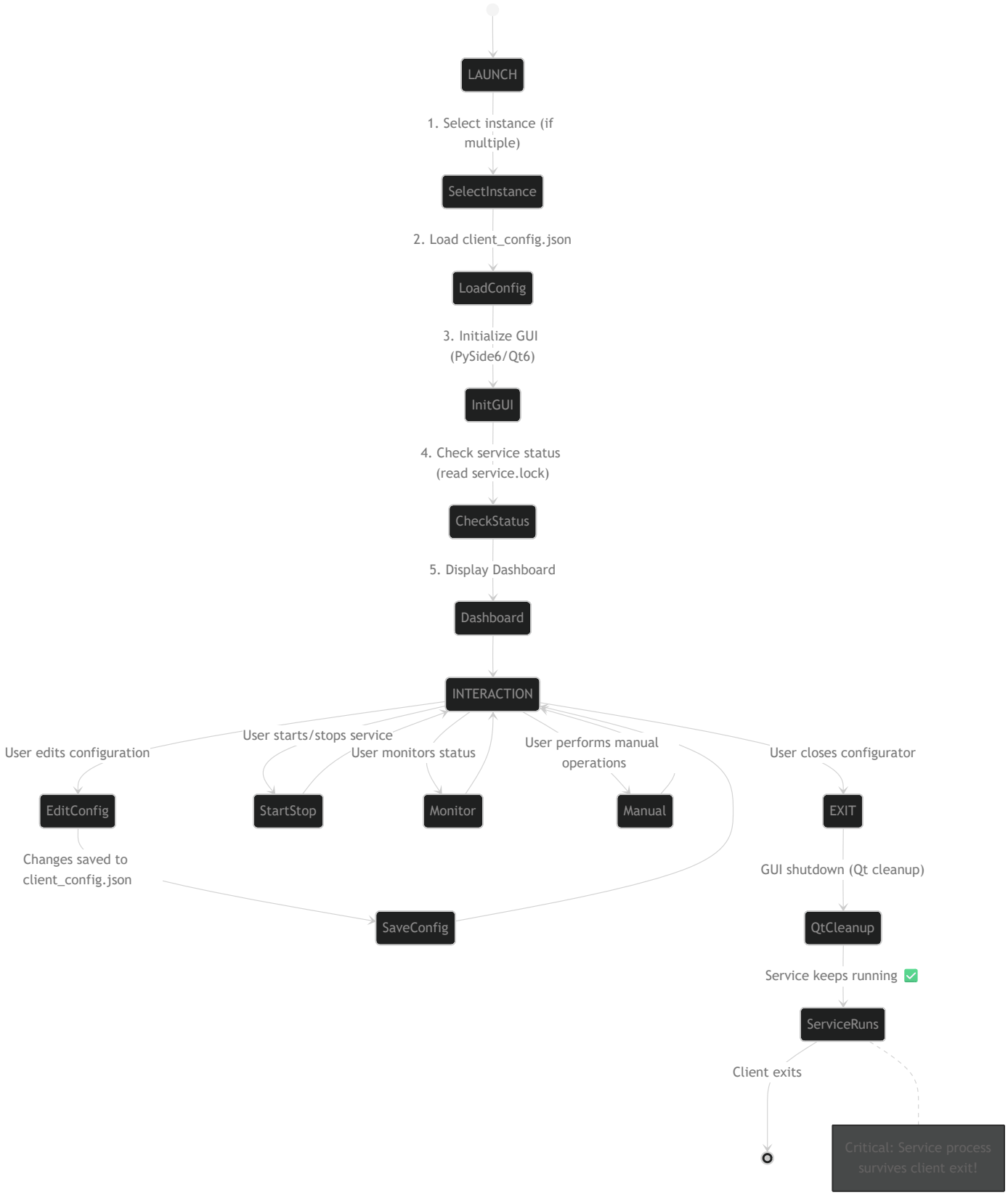
Launchers:

- `run_configurator.py` - GUI launcher with instance selector
- `run_client_a.py` - GUI launcher for “default” instance (test fixture)
- `run_client_b.py` - GUI launcher for “client_b” instance (test fixture)

Components:

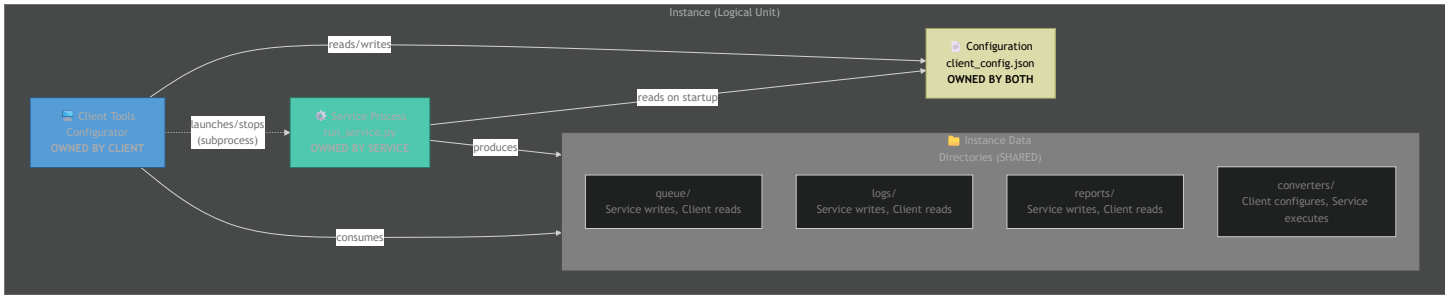
- **Dashboard** - Service status, converter health, statistics
- **Connection** - Server URL, API token, proxy settings
- **Setup** - Station info, location, purpose
- **Converters** - Add/edit/remove file converters
- **Queue** - View pending uploads, manual operations
- **Logs** - Real-time log viewer

Client Lifecycle



Relationship & Ownership

Ownership Model

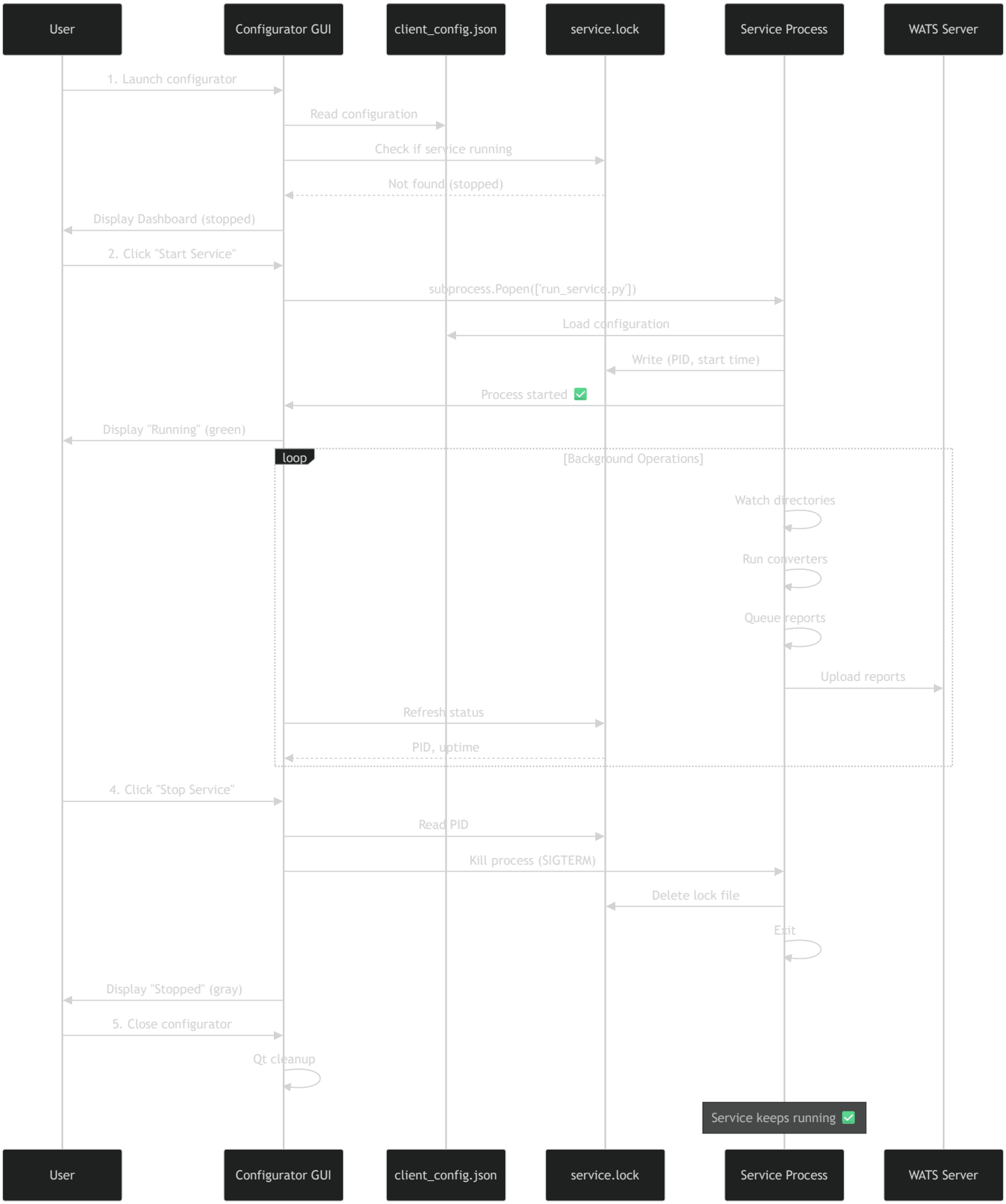


Key Principles

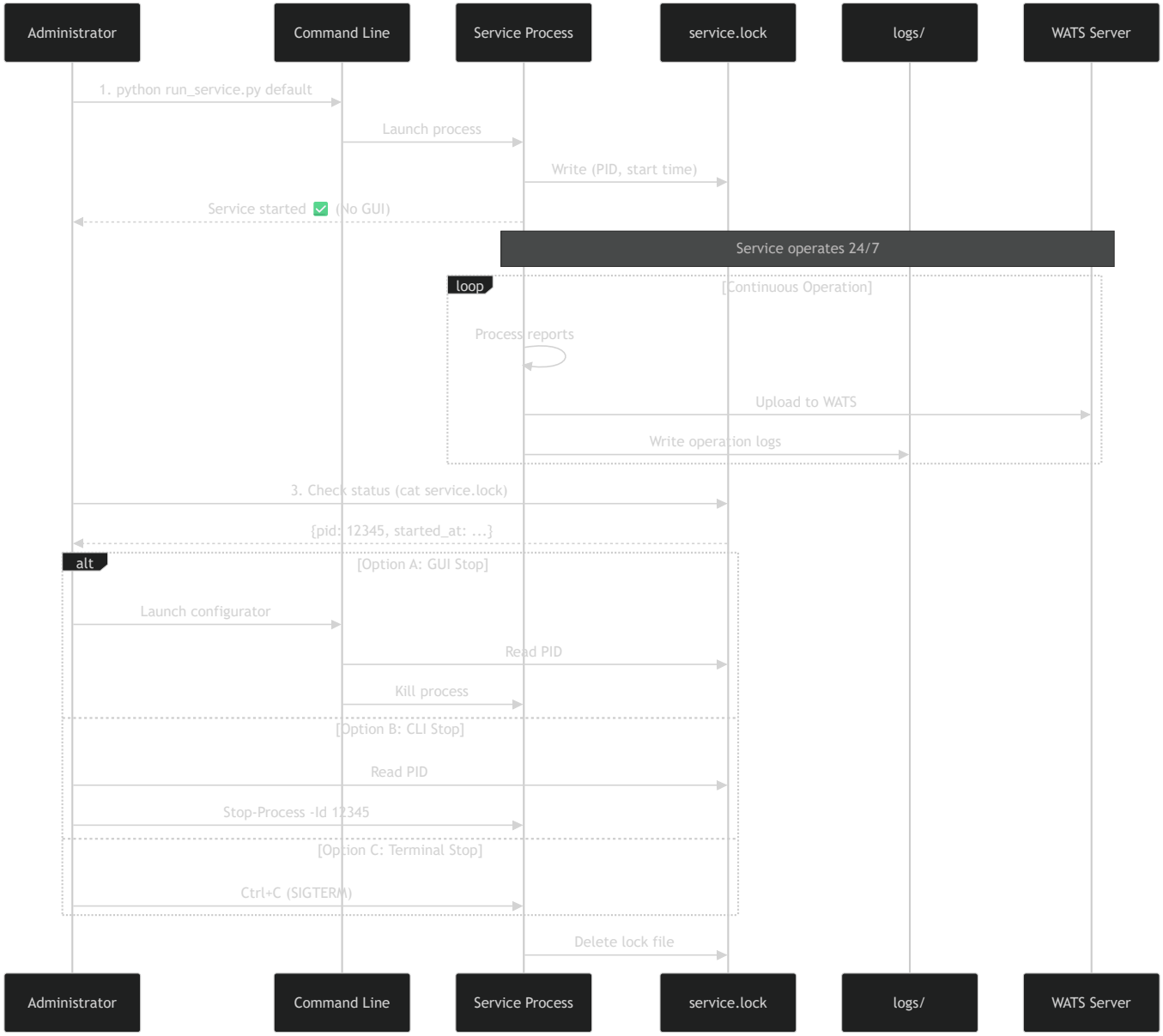
1. **Service is Self-Contained**
   - Service owns its process lifecycle
   - Service executes all background operations
   - Service does NOT depend on client being running
2. **Client is a Controller**
   - Client launches service as subprocess
   - Client reads service status via lock files
   - Client terminates service via PID (OS-level stop)
   - Client does NOT contain service logic
3. **Configuration is Shared**
   - Both read from `client_config.json`
   - Client writes changes, service reads on startup
   - Service does NOT hot-reload config (restart required)
4. **Data Directories are Shared**
   - Service produces data (queue, logs, reports)
   - Client consumes data (display, monitoring)
   - File-based communication (no IPC/sockets required)

How They Work Together

Scenario 1: Normal Operation (GUI + Service)



Scenario 2: Headless Operation (Service Only)



Scenario 3: GUI Crash Recovery

Syntax error in text

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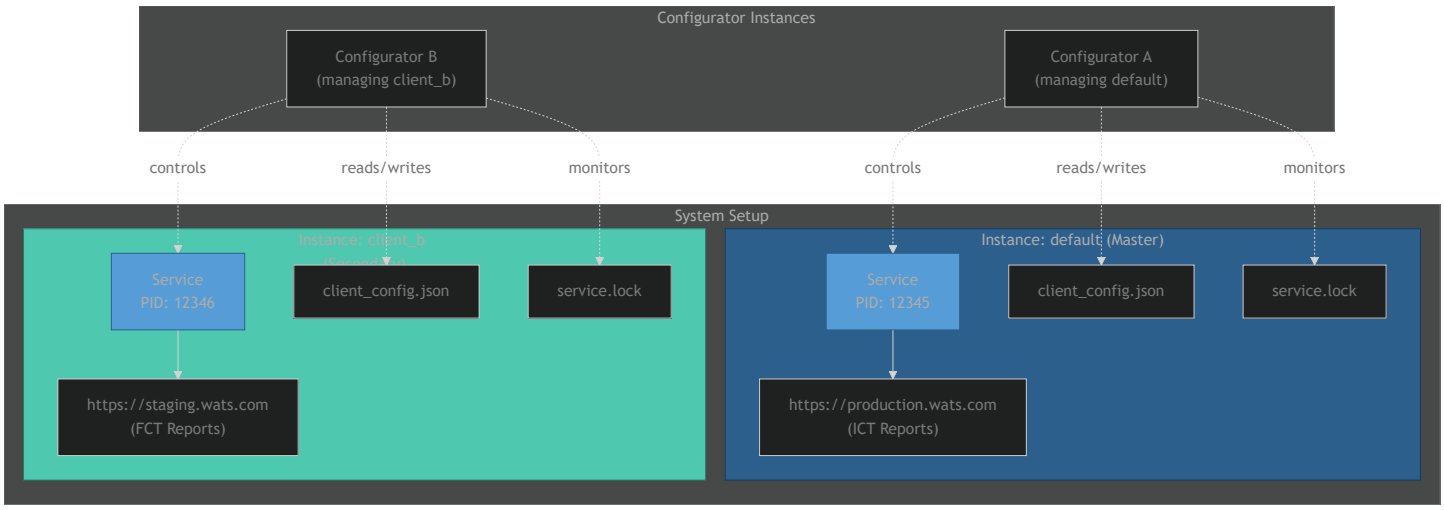
Scenario 4: Multi-Instance Setup



Configurator Workflow:

1. **Launch** `run_configurator.py` → Scans `C:\ProgramData\pyWATS\instances\` → Finds: `default`, `client_b` → Shows dropdown: "Select Instance"
2. **User selects "default"** → Loads `default/client_config.json` → Reads `default/service.lock` → Displays dashboard (service running)
3. **Launch another Configurator** → Select "client_b" → Manages independently ✅ → **Two GUIs managing two services simultaneously**

Communication Methods

Client → Service Communication

Method	Usage	Direction
Subprocess Launch	Start service	Client → Service
OS Process Kill	Stop service	Client → Service
Config File Write	Update settings	Client → Service (on restart)

Service → Client Communication

Method	Usage	Direction
Lock File	PID, uptime, status	Service → Client
Log Files	Detailed operations	Service → Client
Queue Directory	Report status	Service → Client
Reports Directory	Processed reports	Service → Client

Note: No IPC, sockets, or shared memory - communication is entirely file-based for simplicity and reliability.

Configuration Management

Who Edits Config?

Client (Configurator GUI) - Writes to `client_config.json`

- Server URL, API token
- Station name, location, purpose
- Converter definitions (watch folders, executables)
- Proxy settings
- Feature flags

Service (AsyncClientService) - Reads from `client_config.json`

- Loads config on startup (one-time read)
- Does NOT hot-reload (restart required for changes)
- Validates config before starting operations

Config Change Workflow

```
1. User opens Configurator
2. User edits settings (e.g., changes server URL)
3. User clicks "Save" or navigates away
   → client_config.json updated
4. Service keeps using OLD config (no hot-reload)
5. User clicks "Stop Service"
6. User clicks "Start Service"
   → Service reads NEW config ✅
7. Service operates with updated settings
```

Design Decision: No hot-reload prevents mid-operation config changes that could corrupt in-flight uploads.

Service Discovery & Management

Lock File Format

Location: `C:\ProgramData\pyWATS\instances\{instance_id}\service.lock`

Content:

```json
{
  "pid": 12345,
  "instance_id": "default",
  "instance_name": "Client A (Master)",
  "started_at": "2026-02-14T10:30:00.123456",
  "service_url": "https://python.wats.com",
  "config_path": "C:/ProgramData/pyWATS/instances/default/client_config.json"
}
```

Lifecycle:

- Created: When service starts (before operations begin)
- Updated: Never (static metadata)
- Deleted: When service stops gracefully

Stale Lock Detection:

```python
from pywats_client.core import service_manager

# Check if PID is actually running
status = service_manager.get_service_status("default")
if status.is_running:
    print(f"Service running: PID {status.pid}")
else:
    print("Service stopped (stale lock cleaned up)")
```

Service Management API

Python API

```python
from pywats_client.core import service_manager

# Start service (background subprocess)
service_manager.start_service("default", wait=False)

# Stop service (graceful SIGTERM)
service_manager.stop_service("default", force=False)

# Restart service
service_manager.restart_service("default")

# Get status
status = service_manager.get_service_status("default")
print(f"Running: {status.is_running}")
print(f"PID: {status.pid}")
print(f"Uptime: {service_manager.get_uptime(status.started_at)}")

# List all instances
for service in service_manager.list_all_services():
    print(f"{service.instance_id}: {service.is_running}")
```

Command Line

```
# Start service directly
python run_service.py default

# Stop service (via PID from lock file)
$lock = Get-Content C:\ProgramData\pyWATS\instances\default\service.lock | ConvertFrom-Json
Stop-Process -Id $lock.pid

# Check status
Get-Content C:\ProgramData\pyWATS\instances\default\service.lock | ConvertFrom-Json | Format-Lis
```

Deployment Scenarios

Development Environment

Setup:

- Developer runs Configurator for config/testing
- Service runs when needed (manual start/stop)
- Both on same machine

Usage:

```
# Interactive mode
python run_configurator.py
# Click "Start Service" when testing
# Click "Stop Service" when done
```

Production Test Station (Single Instance)

Setup:

- Service runs as Windows Service (auto-start on boot)
- Configurator used for initial setup + troubleshooting only
- Service runs 24/7 unattended

Usage:

```
# Initial setup (one-time)
python run_configurator.py
# Configure server URL, station name, converters
# Close configurator

# Install as Windows Service (via NSSM or pywin32)
nssm install pyWATS "C:\Python\python.exe" "C:\pyWATS\run_service.py" "default"
nssm start pyWATS

# Service runs forever (survives reboots)
```

Production Test Station (Multi-Instance)

Setup:

- Multiple services running (e.g., default, client_b)
- Each instance uploads to different WATS server or process
- Services run as separate Windows Services

Usage:

```
# Instance 1: ICT uploads to Production
nssm install pyWATS_ICT "C:\Python\python.exe" "C:\pyWATS\run_service.py" "default"

# Instance 2: FCT uploads to Staging
nssm install pyWATS_FCT "C:\Python\python.exe" "C:\pyWATS\run_service.py" "client_b"

# Start both
nssm start pyWATS_ICT
nssm start pyWATS_FCT

# Configure each via Configurator (select instance from dropdown)
python run_configurator.py  # Select "default"
python run_configurator.py  # Select "client_b"
```

Troubleshooting

Service Won't Start

Check:

1. Config file exists: `C:\ProgramData\pyWATS\instances\{instance_id}\client_config.json`
2. Required directories exist: `queue/` , `logs/` , `reports/` , `converters/`
3. No stale lock file (or delete it)
4. Check logs: `C:\ProgramData\pyWATS\instances\{instance_id}\logs\`

Fix:

```
# Delete stale lock file
Remove-Item C:\ProgramData\pyWATS\instances\default\service.lock -ErrorAction SilentlyContinue

# Recreate directories
python -c "from pathlib import Path; (Path('C:/ProgramData/pyWATS/instances/default/queue')).mk
```

Configurator Shows Wrong Status

Issue: Service is running but GUI shows "Stopped"

Cause: Stale UI state (not refreshed)

Fix: Click "Refresh" button on Dashboard

Service Crashes After GUI Exit

Issue: Service dies when Configurator closes

Cause: Using old architecture (in-process service)

Fix: Update to latest version (0.2.0+) with subprocess-based service

Verify:

```
# Start service via GUI
python run_configurator.py
# Click "Start Service"

# Check service is subprocess (not thread)
Get-Process python | Where-Object {$_.Id -eq $lock.pid}
# Should show separate process, not child of GUI
```

Best Practices

For Operators

1. **Use GUI for Setup** - Configure via Configurator, then close it
2. **Service Runs Unattended** - Don't keep GUI open 24/7
3. **Restart Service After Config Changes** - Changes require service restart
4. **Monitor via Dashboard** - Launch Configurator to check status anytime
5. **Check Logs for Errors** - Logs directory contains detailed diagnostics

For Administrators

1. **Install as Windows Service** - For production reliability (auto-start)
2. **Use System-Wide Paths** - `C:\ProgramData\pyWATS` for multi-user access
3. **Separate Instances for Isolation** - Different test processes = different instances
4. **Back Up Config Files** - `client_config.json` contains all settings
5. **Monitor Lock Files** - Detect crashed services (stale locks)

For Developers

1. **Never Run Service In-Process** - Always use subprocess (architecture rule)
2. **Service is Headless** - No GUI dependencies allowed
3. **Lock Files for IPC** - File-based communication only
4. **Config is Read-Only for Service** - No hot-reload design
5. **Test Service Independence** - Service must survive client exit

Summary: Service vs Client

Aspect	Service	Client
Purpose	Process test reports, upload to WATS	Configure and monitor services
Lifecycle	Long-running (24/7)	Transient (start/stop as needed)
User Interaction	None (headless)	Full GUI (PySide6/Qt6)
Dependencies	Minimal (core libraries only)	GUI frameworks (PySide6)
Process Model	Independent subprocess	Parent/controller process
Crash Behavior	Auto-recovers, keeps running	Can crash without affecting service ✅
Configuration	Read-only (load on start)	Read-write (edit and save)
Data Production	Produces (queue, logs, reports)	Consumes (displays data)
Critical Path	YES (core mission)	NO (management only)
Deployment	Install as Windows Service	Optional tool (not always installed)

Related Documentation

- Installation Guide - Initial setup
- Configurator Guide - GUI walkthrough
- Multi-Instance Setup - Running multiple services
- Converter Development - Creating custom converters
- API Reference - Python API documentation

Key Takeaway: The **Service** does the work (background processing), the **Client** manages the work (configuration and monitoring). They are **decoupled** for reliability - the service runs independently and survives client crashes.